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# Medical patient boarding in the emergency department as a source of crowding and delay-related harm, impacting patient outcomes and the efficiency of urgent and emergency care

Nicholas Howlett ,[1] James Cameron ,[2] Richard Wood [1,3]

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[1]NHS Bristol North Somerset and South Gloucestershire ICB, Bristol, UK
[2]Emergency Department, North Bristol NHS Trust, Bristol, UK
[3]School of Management, University of Bath, Bath, UK

**Correspondence to**
Dr James Cameron; james.cameron@nbt.nhs.uk

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## ABSTRACT

**Background** Previous studies have indicated that crowding within the emergency department (ED) is associated with longer lengths of stay in the ED and higher mortality. Boarding, the time patients spend waiting for an inpatient bed after ED assessment, represents a clinically unproductive delay and occupies scarce ED resources. We explore the association of medical patient boarding not only with their outcomes but also indirectly for other patients in the ED and those in and awaiting ambulances.

**Methods** A retrospective cohort study using routine data for 3 EDs in England from July 2023 to May 2025 was performed. Direct, delay-related harm outcomes of medical patient boarding time were investigated: inpatient length of stay and 30-day re-admission and mortality rates. Indirect, crowding-mediated outcomes of medical patient boarding levels consisted of time in ED for non-admitted patients, ambulance handover times and Category 1, Category 2 and Category 3 ambulance response times. Mixed-effects regression analysis modelled each relationship while controlling for potential confounding.

**Results** The study examined 49 034 medical admissions, 210 334 non-admitted ED patients, 88 633 ambulance handovers and 146 278 ambulance responses. Medical patients, accounting for two-thirds of ED admissions, constituted 81% of total ED boarding time. Regression analysis showed that for a typical 25-bed ED, each additional five medical boarders was associated with an extra 7.9 min and 51.9 min ambulance response times for Category 2 and Category 3 calls and an extra 9.9 min of extra ambulance handover time. For admitted medical patients, each additional 4 hours of boarding time was associated with an extra 8.6 hours of inpatient length of stay and an 8.4% increase in the odds of 30-day mortality.

**Conclusion** Boarding medical patients in the ED is associated with a significant increase in their risk of harm, including higher mortality and longer hospital stays. It also adversely affects other patients by delaying ambulances and reducing overall bed availability through extended inpatient stays.

### WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ A prolonged length of stay in the emergency department (ED) is known to be associated with increased mortality for those patients.

### WHAT THIS STUDY ADDS

⇒ Delays to medical admission, which constitute 81% of boarding time in the ED, are associated with increased mortality, re-admission and hospital length of stay.

⇒ Consequences extend to other patients through delayed ambulance offload and ambulance response times.

### HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ Boarding medical patients in the ED is associated with a longer inpatient length of stay and therefore fewer beds available for future patients.

⇒ Stopping this practice would likely free up much needed hospital capacity, improve ambulance response times and reduce mortality.

## INTRODUCTION

Boarding is the practice of holding patients in the emergency department (ED) while waiting for an inpatient bed. Boarding may occur in designated cubicles or in less appropriate escalation areas such as corridors. Patients subject to boarding are predominantly awaiting admission to a medical ward (Acute Medicine and Geriatrics specialties).[1]

ED crowding is a significant problem in many countries.[2 3] Causes may be broadly classified as: input (too many or complex patient arrivals), throughput (constraints in staffing, diagnostics, ED beds and productivity) and output (limited inpatient bed availability, often worsened by insufficient intermediate and social care capacity).[4 5] Consequences of ED crowding include delayed ambulance handover times.[6]

Boarding represents a critical output issue, with delays in transferring patients to inpatient wards lengthening their time spent in the ED. Medical patients needing admission are often older and frail, requiring cubicles in the majors and resuscitation areas of the ED. Crowding here will particularly affect ambulance offloads into the ED, in effect taking ambulances off the road and delaying response times for ambulances to get to new patients.

Delay-related harm refers to any adverse health outcome experienced by patients due to prolonged

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ED length of stay before admission to an inpatient hospital ward. These harms include mortality[5–8] and prolonged inpatient length of stay.[9–11]

This study seeks to quantify the association between medical patient boarding and six indicators of patient harm covering ED input, throughput and output. The hypothesis is that the boarding of medical patients is associated with indirect harm to other patients via crowding as well as direct delay-related harm to those experiencing boarding themselves.

## METHODS

### Setting

This study was conducted within the Bristol, North Somerset and South Gloucestershire (BNSSG) Integrated Care System in South-West England. BNSSG serves a one-million-person population across diverse communities and with an age profile similar to other core cities in England.[12] The region includes three hospitals with EDs: Southmead Hospital (20 beds including 6 resuscitation beds), Bristol Royal Infirmary (24 beds including 8 resuscitation beds) and Weston General Hospital (13 beds including 4 resuscitation beds).

### Study design

This was a retrospective observational data-linkage cohort analysis in which the multiple considered outcomes were examined separately. In each such instance, the approach involved statistically modelling the level of association between the relevant boarding-related exposure and the outcome, while controlling for the effect of other possibly confounding variables. Linear regression was used for continuous outcomes and logistic regression was used for binary-valued outcomes. The STROBE (Strengthening Reporting of Observational Studies in Epidemiology) checklist was followed within this study.

### Exposures

There are two distinct types of exposures that result from boarding: delay and crowding. Medical boarders themselves, by definition, experience delays between the decision to admit and their admission, which is measured in continuous time. Other emergency care patients (who have not yet been admitted) experience the impact of boarding through crowding of the ED, which is measured as the number of ED patients currently waiting for a medical bed at the time of their arrival, ambulance handover or dispatch.

### Outcomes of interest

This study investigated six outcomes, three outcomes reflecting delay-related impacts and three reflecting crowding-related impacts. The delay-related outcomes, measured for those who have directly experienced boarding, were inpatient length of stay, 30-day re-admission risk and 30-day mortality risk. The crowding-related outcomes, measured for those potentially indirectly impacted by boarding, were ambulance response time for all patients arriving by ambulance, ambulance handover time and ED lengths of stay for non-admitted patients. Ambulance response time is defined as the interval between when the 999 call is answered and when the first attending ambulance is recorded as arriving at the patient's location. Ambulance handover time is defined as the interval between the ambulance's arrival at the ED and the completion of the clinical handover and physical transfer to the ED clinical staff. ED length of stay is defined as the interval between the patient's arrival time at the ED and their time of discharge (departure) from the department.

### Participants

This study employed multiple cohorts, each defined by the specific outcome under investigation. Participants were eligible if their outcome occurred between 1 July 2023 and 1 May 2025. For the delay-related outcomes, the participants were those requiring admission from the ED to the medical specialties. Admissions after 1 April 2025 are excluded, allowing a 30-day follow-up period for re-admission and mortality. Admissions were excluded if the boarding delay exceeded 24 hours (this was due to concerns around these cases relating to a different patient group, as these extended waits were likely driven by delayed referrals for social care rather than acute medical issues). For the crowding-related outcomes, the participants are those requiring an ambulance response, those requiring an ambulance handover and those attending ED but not requiring admission. Any participants with missing data were excluded from the analysis.

### Statistical analysis

All regression models included a standard set of covariates to account for confounding: patient age (discretised into decades to address non-linearity), sex, deprivation decile (based on the Index of Multiple Deprivation), event time (6-hour bands), event month and weekend occurrence. Outcome-specific confounders were also incorporated. For crowding-related outcomes, non-medical ED occupancy at the event time was included as a confounder to isolate crowding attributable to medical boarders. For delay-related outcomes, the following confounders were included: patient comorbidity score (calculated using the Van Walraven Elixhauser method[13] from admission International Classification of Diseases ICD-10 codes); ED attendances within 60 days prior to admission (excluding the index attendance); time from arrival to decision to admit, to isolate time being seen and treated in the ED before that decision; and, in providing a proxy for admission delays due to exit block from medical wards, the number of admitted medical patients who remain admitted but are ready for discharge (in the National Health Service, this is known as having 'No Criteria to Reside' (NCtR)).

To account for the non-independence of observations arising from multiple events (admission, ambulance handover, etc) per patient, all regression analyses employed mixed-effects models, which included a random intercept for each patient (clustering by unique patient identifier). All models were estimated using the R programming language (V.4.5.1) and the lme4 package.[14] To account for variation in ED size, all patient counts (medical boarders, non-medical ED occupancy and patients with NCtR) were normalised as proportions of designated ED bedded capacity (majors and resuscitation beds). Quadratic terms were included for continuous variables to capture potential non-linear relationships.

From the fitted regression models, effect sizes and values of p were derived. For interpretability, crowding-related effects were expressed in terms of each additional 20% increase in medical patients as a proportion of designated ED bedded capacity, and delay-related effects in terms of each additional 4-hour increase in medical boarding time. The 20% increase was chosen as a significant step in decreased available capacity and crowding, and as a marker that was applicable between departments of different size.

Additionally, to examine the relationship between crowding due to boarding and the crowding-related outcomes, we employed a standardisation approach to produce a confounder-controlled descriptive summary of our data. The primary goal was to better visualise the relationships, assess for potential non-linearities

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(ie, effects that worsen dramatically at high crowding levels) and facilitate comparison to national performance targets.[15–17] The standardisation process was designed to isolate the effect of crowding by removing the influence of known confounders (eg, patient age, arrival time, etc) from the outcome variable. For each outcome, we first ran a linear regression model using only the confounders, deliberately excluding the crowding-related exposure variable. These models allowed us to calculate a standardised observation for every patient. This value was calculated as the cohort-level empirical mean plus the residual between the patient's actual outcome and the outcome predicted by the confounder-only model. The resulting value essentially represents the patient's actual outcome minus the portion of the outcome that was explained by the confounders, with the final observations centred around the overall population mean. Boxplots were produced for these standardised outcomes, stratified by the level of ED crowding at the event time (expressed as a percentage of designated ED bedded capacity, binned into 21 bands of 5% width from 0% to 105%, covering 95% of the data).

Due to insufficient event counts, a similar approach was not feasible for delay-related outcomes. Instead, associations between boarding time of medical patients and outcomes were illustrated using conditional effects plots.[18] To contextualise the relationship between boarding delays and 30-day re-admission and mortality, additional regression models were constructed with admission delays dichotomised at 4 hours, allowing for the calculation of numbers needed to harm for boarding delays exceeding 4 hours relative to those below.

### Data sources and definitions

Data on ambulance response and handover times were sourced from the Ambulance Data Set.[19] ED data were sourced from the Emergency Care Data Set.[20] Admitted patient length of stay (LOS) and comorbidity measures were sourced from the Secondary Uses Service Admitted Patient Care data set, a derivative of the Hospital Episode Statistics data set.[21] Mortality data were sourced from the Civil Registrations of Death data set.[22] Finally, data on the ED bedded capacity and numbers of patients with NCtR were provided directly by the hospitals. Where applicable, the data sources were linked via a pseudonymised patient identity and event time stamps. All data were routinely collected and anonymised with no patient identifiable information included, and so no specific ethics permissions were required. All other necessary permissions were obtained for use of the data in the study.

## RESULTS

### Study cohorts

All analyses on delay-mediated outcomes were performed using the same cohort of medical admissions. A total of 52 950 medical admissions from the ED were initially identified within the study period. From these, 1444 patients (2.7% of total) were excluded due to boarding time exceeding 24 hours and an additional 2472 cases (4.7% of the total) were excluded due to missing data. For the analyses of crowding-mediated outcomes, the separate cohorts comprised ambulance responses, ambulance handovers and ED attendances for non-admitted patients within the study period. For the analysis involving ambulance response times, cases were categorised by response type, where 21 789 (Category 1, representing the most critical incidents), 108 248 (Category 2) and 36 832 (Category 3) were identified within the study period. Cases were subsequently excluded across these groups due to missing confounders, specifically 4190 in Category 1 (19.2% of total), 14 035 in Category 2 (12.9%) and 3366 in Category 3 (9.1%). This left 18 599 (Category 1), 94 213 (Category 2) and 33 466 (Category 3) ambulance responses for inclusion in the study. Of the 99 358 identified ambulance handovers, 10 725 (10.7%) were excluded due to missing confounders, leaving a final sample of 88 633. Finally, from the 215 618 ED arrivals identified during the study period, 5284 were excluded due to missing data (2.4% of the total), resulting in 210 334 ED arrivals considered for inclusion. A STROBE flow chart representation of the data exclusions is provided in the online supplemental figure SM1. Cohort sizes, demographics and associated confounding variables for all outcomes are presented in table 1. The distribution of outcomes stratified by patient demography is presented visually in the online supplemental figures SM2 and SM3.

### Sample characteristics

The mean time to decision to admit for medical patients across the three hospitals was 189 min (3.15 hours). Medical patients experienced substantially longer boarding times—a mean of 374 min (6.2 hours) compared with 176 min (2.9 hours) for other patients. As medical patients accounted for two-thirds of ED admissions, they contributed to 81% of the total admission delay time. The distribution of medical boarding times was markedly right-skewed: 44% of boardings occurred within 0–4 hours, followed by 27% within 4–8 hours, 14% within 8–12 hours, 8% within 12–16 hours, 5% within 16–20 hours and 3% within 20–24 hours (see online supplemental figures SM2 and SM3 for the full distributions).

### Crowding-mediated impact of medical boarding

Table 2 shows how crowding due to boarding medical patients is associated with delays in ambulance handover, the subsequent delays on ambulance response times and lengths of stay in the ED for non-admitted patients. Of note, the association with Category 1 ambulance response times, although statistically significant, was much less marked than Categories 2 and 3, presumably due to the ambulance service prioritising these more critical patients over others who waited longer. Figure 1 shows in more detail how this association is non-linear: the effect of the first 25% of majors and resuscitation ED beds being occupied by medical boarders is modest, with an increasingly negative effect as the departments become more crowded. Full model details are provided in the online supplemental table SM1.

### Delay-mediated impact of medical boarding

Figure 2 shows the association between medical boarding time and the outcomes of medical patients, with table 3 summarising these associations for each additional 4 hours of boarding. These were most marked for inpatient length of stay (excluding time spent in ED) and mortality. The patients in our study, between 1 July 2023 and 1 April 2025, had a mean boarding delay in the ED of 6.2 hours, which corresponds to an estimated additional 14.6 hours length of stay on the wards. Each additional 4 hours boarding time was associated with an 8.4% (95% CI 5.3% to 11.6%) increase in the odds of mortality within 30 days. The number needed to harm (NNH) was that for every 69 patients boarding more than 4 hours for a medical bed, one additional death would be expected. The NNH for an additional re-admission under the same exposure was 218. Full model details are provided in the online supplemental table SM2.



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**Table 1** Descriptive summary of the cohorts used to investigate crowding-related outcomes

| | | Ambulance response time | | | Ambulance handover time | ED length of stay for non-admitted patients | Inpatient LOS, 30-day re-admission rate, 30-day mortality rate |
|---|---|---|---|---|---|---|---|
| | | Category 1 (most critical) | Category 2 | Category 3 | | | |
| Event description | | Ambulance dispatch | Ambulance dispatch | Ambulance dispatch | Ambulance handover | ED arrival | Admission from ED |
| Number of events | | 18 599 | 94 213 | 33 466 | 88 633 | 210 334 | 49 034 |
| Number of medical boarders, as a percentage of ED bedded capacity (at times of events) | Median (IQR) | 49.5% (33.3% to 67.9%) | 48.4% (32.3% to 66.7%) | 44.6% (28.7% to 61.8%) | 45.0% (25.0% to 65.0%) | 45.0% (25.0% to 65.0%) | – |
| Medical boarding time (hours) | Median (IQR) | – | – | – | – | – | 5.1 (2.3 to 9.5) |
| Age | | | | | | | |
| 29 years old or under | n (%) | 4077 (22%) | 11 705 (12%) | 3123 (9.3%) | 9294 (10%) | 71 945 (34%) | 3056 (6.2%) |
| 30–39 years old | n (%) | 2326 (13%) | 8860 (9.4%) | 2447 (7.3%) | 7819 (8.8%) | 37 002 (18%) | 3227 (6.6%) |
| 40–49 years old | n (%) | 2063 (11%) | 8660 (9.2%) | 2177 (6.5%) | 7610 (8.6%) | 27 218 (13%) | 3556 (7.3%) |
| 50–59 years old | n (%) | 2012 (11%) | 10 890 (12%) | 2840 (8.5%) | 9620 (11%) | 23 679 (11%) | 5310 (11%) |
| 60–69 years old | n (%) | 2053 (11%) | 12 320 (13%) | 3270 (9.8%) | 11 500 (13%) | 18 810 (8.9%) | 6827 (14%) |
| 70–79 years old | n (%) | 2439 (13%) | 15 885 (17%) | 5829 (17%) | 16 157 (18%) | 16 166 (7.7%) | 10 502 (21%) |
| 80–89 years old | n (%) | 2629 (14%) | 18 525 (20%) | 9220 (28%) | 19 169 (22%) | 12 255 (5.8%) | 12 101 (25%) |
| 90–99 years old | n (%) | 963 (5.2%) | 7128 (7.6%) | 4401 (13%) | 7265 (8.2%) | 3190 (1.5%) | 4360 (8.9%) |
| 100 years old or over | n (%) | 37 (0.2%) | 240 (0.3%) | 159 (0.5%) | 199 (0.2%) | 69 (<0.1%) | 95 (0.2%) |
| Sex | | | | | | | |
| Female | n (%) | 9871 (53%) | 50 433 (54%) | 18 771 (56%) | 47 085 (53%) | 108 279 (51%) | 25 177 (51%) |
| Male | n (%) | 8728 (47%) | 43 780 (46%) | 14 695 (44%) | 41 548 (47%) | 102 055 (49%) | 23 857 (49%) |
| Deprivation decile | | | | | | | |
| 1 (most deprived) | n (%) | 3379 (18%) | 13 941 (15%) | 4189 (13%) | 12 776 (14%) | 30 756 (15%) | 6910 (14%) |
| 2 | n (%) | 2230 (12%) | 9847 (10%) | 3057 (9.1%) | 8824 (10.0%) | 22 826 (11%) | 4870 (9.9%) |
| 3 | n (%) | 2014 (11%) | 9969 (11%) | 3437 (10%) | 9187 (10%) | 22 375 (11%) | 5129 (10%) |
| 4 | n (%) | 1977 (11%) | 9855 (10%) | 3686 (11%) | 9296 (10%) | 21 496 (10%) | 4535 (9.2%) |
| 5 | n (%) | 1439 (7.7%) | 7342 (7.8%) | 2961 (8.8%) | 7129 (8.0%) | 15 503 (7.4%) | 4030 (8.2%) |
| 6 | n (%) | 1447 (7.8%) | 7908 (8.4%) | 2909 (8.7%) | 7615 (8.6%) | 18 490 (8.8%) | 4159 (8.5%) |
| 7 | n (%) | 1875 (10%) | 9761 (10%) | 3523 (11%) | 9227 (10%) | 24 031 (11%) | 5560 (11%) |
| 8 | n (%) | 1453 (7.8%) | 8290 (8.8%) | 3166 (9.5%) | 7802 (8.8%) | 20 547 (9.8%) | 4954 (10%) |
| 9 | n (%) | 1391 (7.5%) | 8338 (8.9%) | 3355 (10%) | 8099 (9.1%) | 16 620 (7.9%) | 4460 (9.1%) |
| 10 (least deprived) | n (%) | 1394 (7.5%) | 8962 (9.5%) | 3183 (9.5%) | 8678 (9.8%) | 17 690 (8.4%) | 4427 (9.0%) |
| Time of event | | | | | | | |
| 00:00–05:59 | n (%) | 3031 (16%) | 13 354 (14%) | 5919 (18%) | 18 185 (21%) | 21 105 (10%) | 5297 (11%) |
| 06:00–11:59 | n (%) | 4472 (24%) | 25 755 (27%) | 10 288 (31%) | 16 882 (19%) | 48 741 (23%) | 12 090 (25%) |
| 12:00–17:59 | n (%) | 5468 (29%) | 30 048 (32%) | 9107 (27%) | 26 580 (30%) | 80 536 (38%) | 18 151 (37%) |
| 18:00–23:59 | n (%) | 5628 (30%) | 25 056 (27%) | 8152 (24%) | 26 986 (30%) | 59 952 (29%) | 13 496 (28%) |
| Month of event | | | | | | | |
| January | n (%) | 1736 (9.3%) | 8937 (9.5%) | 3221 (9.6%) | 8095 (9.1%) | 18 175 (8.6%) | 4738 (9.7%) |
| February | n (%) | 1699 (9.1%) | 8739 (9.3%) | 3069 (9.2%) | 7749 (8.7%) | 17 906 (8.5%) | 4325 (8.8%) |

Continued



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**Table 1** Continued

| | | Ambulance response time | | | Ambulance handover time | ED length of stay for non-admitted patients | Inpatient LOS, 30-day re-admission rate, 30-day mortality rate |
|---|---|---|---|---|---|---|---|
| | | Category 1 (most critical) | Category 2 | Category 3 | | | |
| March | n (%) | 1864 (10%) | 9366 (9.9%) | 3385 (10%) | 8781 (9.9%) | 20 397 (9.7%) | 4413 (9.0%) |
| April | n (%) | 885 (4.8%) | 4575 (4.9%) | 1775 (5.3%) | 4354 (4.9%) | 10 142 (4.8%) | 2460 (5.0%) |
| May | n (%) | 976 (5.2%) | 4966 (5.3%) | 1596 (4.8%) | 4567 (5.2%) | 11 178 (5.3%) | 2293 (4.7%) |
| June | n (%) | 925 (5.0%) | 4725 (5.0%) | 1558 (4.7%) | 4307 (4.9%) | 10 205 (4.9%) | 2377 (4.8%) |
| July | n (%) | 1743 (9.4%) | 8904 (9.5%) | 3500 (10%) | 9000 (10%) | 21 828 (10%) | 4755 (9.7%) |
| August | n (%) | 1521 (8.2%) | 8403 (8.9%) | 3750 (11%) | 8831 (10.0%) | 21 706 (10%) | 4726 (9.6%) |
| September | n (%) | 1511 (8.1%) | 8309 (8.8%) | 2998 (9.0%) | 7916 (8.9%) | 20 919 (9.9%) | 4621 (9.4%) |
| October | n (%) | 1856 (10.0%) | 9316 (9.9%) | 2804 (8.4%) | 8430 (9.5%) | 20 978 (10.0%) | 4792 (9.8%) |
| November | n (%) | 1859 (10.0%) | 8961 (9.5%) | 2942 (8.8%) | 8306 (9.4%) | 19 066 (9.1%) | 4727 (9.6%) |
| December | n (%) | 2024 (11%) | 9012 (9.6%) | 2868 (8.6%) | 8297 (9.4%) | 17 834 (8.5%) | 4807 (9.8%) |
| Event on weekend | n (%) | 5457 (29%) | 27 710 (29%) | 10 452 (31%) | 26 682 (30%) | 60 633 (29%) | 13 064 (27%) |
| Number in ED (excluding medical boarders) as a percentage of bedded ED capacity (at times of events) | Median (IQR) | 83.8% (58.3% to 109.2%) | 83.4% (57.9% to 109.0%) | 81.0% (55.0% to 107.1%) | 90.0% (55.0% to 125.0%) | 90.0% (55.0% to 130.0%) | – |
| Time to decision to admit (hours) | Median (IQR) | – | – | – | – | – | 2.78 (1.34 to 4.38) |
| Number of patients with 'No Criteria To Reside' as a percentage of cubicle capacity (at time of event) | Median (IQR) | – | – | – | – | – | 366.7% (325.0% to 876.9%) |
| Van Walraven Elixhauser Comorbidity Score | Median (IQR) | – | – | – | – | – | 5 (0 to 12) |
| Number of ED attendances within the past 60 days | Median (IQR) | – | – | – | – | – | 0.00 (0.00 to 0.00) |

ED, emergency department.



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**Table 2** Crowding-related effect sizes due to increase in the number of medical patients boarding in the ED equal to 20% of the number of majors and resuscitation beds (equal to five additional medical boarders if you have 25 majors and resuscitation beds), including 95% CIs

| Exposure | Outcome | Effect size (95% CI) | P value |
|---|---|---|---|
| + 20% increase in the number of medical boarders, as a % of ED majors and resuscitation capacity | Ambulance response time Category 1 (most critical) | 0.1 (0 to 0.2) min additional response time | 0.002 |
| | Ambulance response time Category 2 | 7.9 (7.7 to 8.1) min additional response time | <0.001 |
| | Ambulance response time Category 3 | 51.9 (49.9 to 54) min additional response time | <0.001 |
| | Ambulance handover time | 9.9 (9.6 to 10.2) min additional handover time | <0.001 |
| | ED length of stay for non-admitted patients | 11.7 (11.2 to 12.2) min additional time in the ED | <0.001 |

Values of p indicate the level of statistical evidence in the associations.
ED, emergency department.

## DISCUSSION

This study contributes to the existing literature by quantifying the impact of medical patient boarding in the ED on both direct and indirect outcomes across the urgent and emergency care pathway. It shows that the boarding of patients admitted under Acute Medicine and Geriatrics, accounting for 81% of total boarding time in the ED, is associated with direct and indirect consequences for these patients and others within the ED and in and awaiting ambulances.

Crowding-related outcomes were standardised to ED size by modelling the effect of medical boarders equivalent to 20% of majors and resuscitation cubicles. For a typical 25-cubicle ED, this equates to five medical boarders. Each additional 20% increase was associated with a 10 min delay in ambulance handover, a

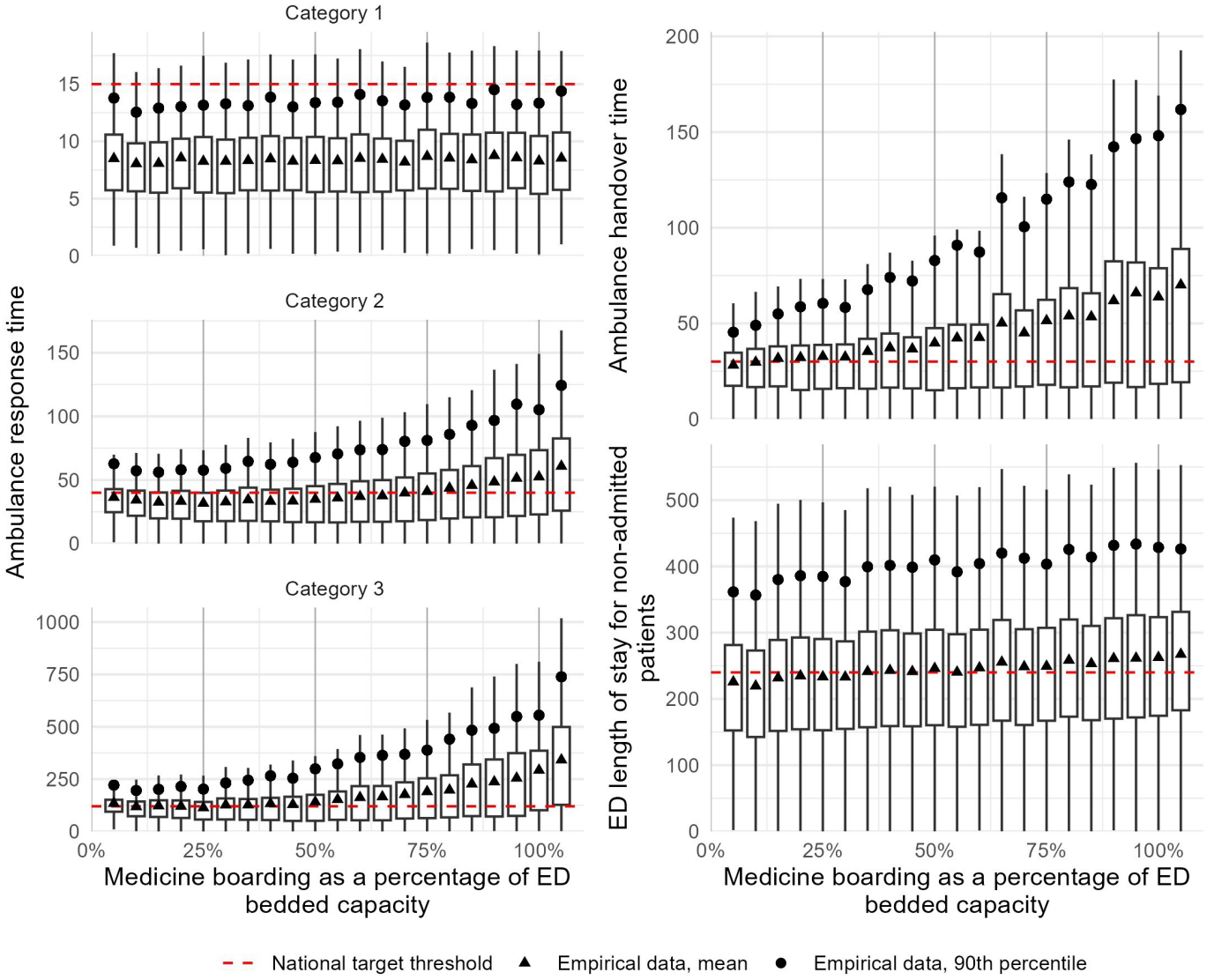


**Figure 1** For crowding-related/indirect impacts, boxplots describing the distribution of outcome values (standardised for confounders and all in minutes) for different levels of medical boarding, as a percentage of emergency department (ED) bedded capacity. The box height represents the IQR, and whiskers are ±1.5 IQR.

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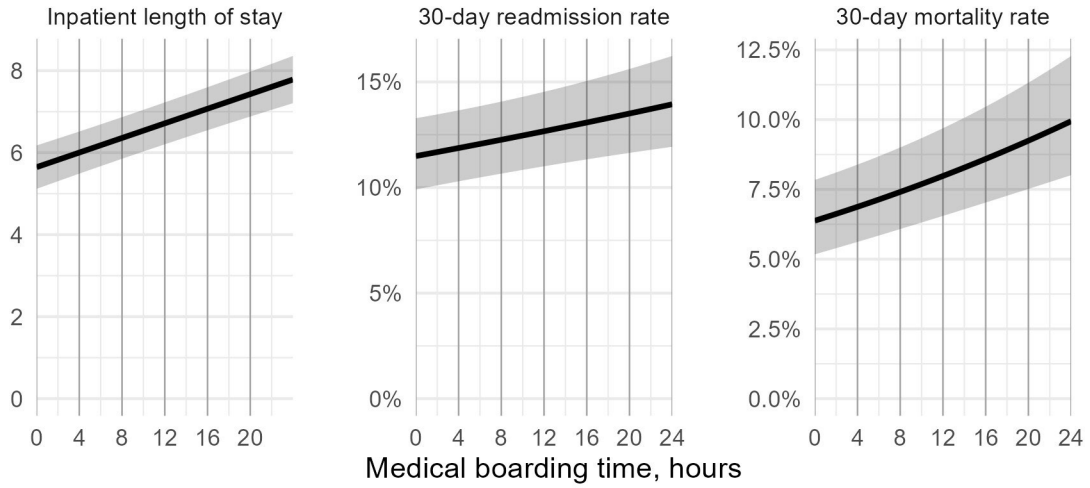


**Figure 2** Delay-related harms of medical boarding in emergency department (ED) on those patients: predicted mean (solid line) and 95% CIs on the predicted mean (shaded areas) based on different durations of boarding time (in hours). Note that inpatient length of stay is measured in days.

**Table 3** Delay-related effect sizes due to increase of medical boarding delay by 4 hours, including 95% CIs

| Exposure | Outcome | Effect size (95% CI) | P value |
|---|---|---|---|
| Each additional 4 hours boarding time for medical patients | Inpatient length of stay | 8.6 (6.9 to 10.2) hours additional LOS | <0.001 |
| | 30-day re-admission rate | 3.8% (1.5% to 6.1%) increase in odds of re-admission within 30 days | <0.001 |
| | 30-day mortality rate | 8.4% (5.3% to 11.6%) increase in odds of mortality within 30 days | <0.001 |

Values of p indicate the level of statistical evidence in the associations.

52 min increase in Category 3 ambulance response time, and a 12 min increase in ED length of stay. Category 1 ambulances—reserved for the most critical patients—were minimally affected, suggesting prioritisation mechanisms remain effective for the highest-acuity cases.

This study quantified delay-related harm for medical boarders. Each 4-hour increase in medical patient ED boarding time was associated with an increase in inpatient length of stay of 8.6 hours. Boarding delays were also associated with increased re-admission and mortality. Each 4-hour delay was associated with an 8.4% increase in the odds of 30-day mortality. When dichotomising our exposure to delays above and below 4 hours of boarding, the NNH (30-day mortality) was 69, which, when considering the mean 3.15-hour time from arrival to decision to admit in our study, is consistent with the result of 72 for overall ED stays exceeding 8 hours as reported in a previous large study.[8] In our sample, 59% were delayed beyond 4 hours, equivalent to eight excess deaths per 1000 admissions. This figure represents 11% of the 63 deaths per 1000 admissions observed. Note additionally that the overall mortality in our study was similar to another study of emergency admissions,[23] suggesting that our studied cohort was broadly representative.

The exclusive focus on medical patient boarding, the integration of specific confounding variables and the accounting of multiple attendances is novel to this study. This approach helped to isolate medical boarding's distinct impact from other sources of ED delays and crowding, such as effects due to exit block from the hospital. The outcomes studied—ambulance response and handover times, ED and inpatient length of stay, re-admission and mortality—extend beyond the immediate ED episode and reflect broader system-level consequences. This coverage extends beyond previous studies, which consider either just mortality[5–8] or prolonged inpatient length of stay.[9–11] Note also that some improvements are made on methodologies previously employed, for example, as suggested in previous studies,[8] we have used mixed-effects models to appropriately capture multiple admissions.

However, several limitations should be acknowledged. The analysis was constrained by the confounders available in the data set. For instance, unavailability of the National Early Warning Score and Rockwood Clinical Frailty Scale precluded a more comprehensive control for patient condition (note that triage criteria data were available but were found to not offer much explanation, at least when considered alongside the Elixhauser Score). Further limitations are the potential effects of missing data and any inconsistencies in the recording of event timestamps.

This study has important implications for practice. The associations observed suggest that reducing medical boarding in the ED could improve ambulance response times, reduce inpatient length of stay, lower re-admission and mortality rates, and ease pressure on elective services—highlighting the potential for significant system-wide gains through boarding reduction. It may also reduce costs and improve patient experience. Current escalation thresholds may be set too high, allowing harmful delays to become normalised. A shift towards proactive boarding minimisation is warranted.

Further research should explore the relationship between ED boarding and subsequent social care needs, including increased support at home or transitions to residential care. The impact of ambulance delays on patient outcomes—particularly for those ultimately admitted—also warrants investigation. Finally, the models developed here could be applied to routine ED data to estimate the cumulative impact of boarding on key outcomes, supporting real-time quality improvement efforts.

**Contributors** All authors contributed to the overall structure, wording and approach of the paper. NH established data linkage for the data sets and subsequent



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analysis of how boarding relates to outcomes. JC proposed boarding of medical patients as an area of focus and contributed to outcomes for analysis. JC is the guarantor. RW initially proposed the analytical study relating to ED data and contributed to the analytical methodology.

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**Patient consent for publication** Not applicable.

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**ORCID iDs**
Nicholas Howlett https://orcid.org/0009-0003-1568-8047
James Cameron https://orcid.org/0000-0003-1254-9669
Richard Wood https://orcid.org/0000-0002-3476-395X

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